

Aradhya Khandelwal

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AI engineer for LLM products: the only engineer on a multi-agent platform behind a managed marketing service, and author of Hypercharge, an open-source toolkit for Claude Code and Cursor agents. Two papers and a patent application.

Skills

LLMs & Agents: Multi-agent systems, LangChain, LangGraph, CrewAI, RAG pipelines, vector databases, tool use, prompt iteration, adversarial critics, OpenRouter (Claude, GPT, Gemini, Llama, DeepSeek), vision models, evals, cost tracking

Backend: Python, FastAPI, async SQLAlchemy, PostgreSQL, Alembic, Postgres job queues, JWT and API-key auth, multi-tenant scoping, pytest, Docker, Render, AWS (EC2, S3), GitHub Actions

Machine Learning: PyTorch, CNNs (1D/2D/3D), ablation studies, synthetic data generation, sim-to-real transfer, scikit-learn

Web: TypeScript, React, Next.js, Playwright

Experience

Co-founder & CTO, The Algorithm Lab (early-stage, not yet incorporated)

Jun 2026 – Present

- Only engineer at an AI marketing startup that runs a managed marketing service. Building the product that automates that service, now in beta.
- Built 7 platform-writer agents, a topic planner, an adversarial critic and a 4-stage campaign planner, all on one company audit.
- Built the website-audit pipeline: HTML extraction, headless Playwright screenshots, Claude vision grading and PDF reports.
- Designed a provider layer on OpenRouter (Claude primary, Gemini as a per-workload fallback) with per-call token and cost tracking. FastAPI, async SQLAlchemy with PostgreSQL and React 19 on Render; 48 API routes, 594 automated tests.

Deep Learning Trainee, DeepTech Division — SSS Defence, Bangalore

Mar 2026 – Present

- Built an acoustic drone detector for a multi-microphone array, trained only on synthetic data. On 40 real field flights it caught 89.4% of drone segments (95% CI 87.7–90.9%); synthetic AUROC 0.995.
- Built the synthetic data pipeline, then diagnosed the model with branch knock-out ablations and hard negatives that cut machinery look-alike scores from 0.79 to 0.04–0.16 on synthetic tests.
- Built a CUDA signal-processing front-end and ported the detector to NVIDIA Jetson Orin with runtime tests and benchmarks.

Automation Intern, Intact Group, Gurgaon

Jul 2025 – Mar 2026

- Built Playwright/Python automation for 25+ regression workflows, cutting manual QA time by ~40%, and wired automated validation into CI/CD during the consolidation of 3 subsidiary firms.

Research Intern, IIT Roorkee

Jun 2025 – Jul 2025

- Built 1D/2D/3D CNNs for 3-class PolSAR and multispectral classification on 11-band satellite data, with band-level ablations. Co-authored the IEEE InGARSS 2025 paper below.

Projects

Hypercharge — toolkit for AI coding agents | [\[GitHub\]](#)

Jun 2026 – Jul 2026

- Open-source Python CLI that wires a repo for Cursor and Claude Code agents: a tree-sitter code graph they query before editing, session memory with SQLite full-text search, and advisory safety hooks; 260 automated tests, MIT licence.

Earmark — in-browser personal voice isolation for voice agents | [\[GitHub\]](#) *in progress*

Sep 2026 – Present

- Causal 10 ms-hop model that keeps only the enrolled voice and gates barge-ins, so an agent is interrupted only by its own user; C++17 engine compiled to WebAssembly.

Review Insights — LLM review analytics | [\[GitHub\]](#)

Jan 2026 – May 2026

- Next.js and PostgreSQL app with Google sign-in and Google Business Profile sync; an LLM pipeline sorts reviews into 6 subject buckets with health scores, tested on 150 real reviews.

SwiftShare — anonymous encrypted file sharing | [\[GitHub\]](#)

Feb 2025 – May 2025

- FastAPI and S3 service with AES-256-GCM, optional passwords and expiring links; basis of my ICDSA paper and patent.

Publications & Patent

Beyond Nonce Uniqueness: A Cryptographic Evaluation of AES-GCM in Secure Anonymous File Sharing Systems

A. Khandelwal, L. Upadhyay, S. Shringi. *ICDSA 2025*, Springer, pp. 323–334. [\[Springer\]](#)

Unsupervised Approach for Classifying Volume Scattering for Surface Roughness and Potential Water-Ice Deposits

S. Muduli, A. Khandelwal, A. K. Maurya, S. K. Singh, D. Singh. *IEEE InGARSS 2025*, pp. 594–598. [\[IEEE Xplore\]](#)

A Secure Anonymous File Sharing System with Optional Password-Based Encryption and Expiry Controls

Indian Patent Application No. 202511066102 A, published 25 Jul 2025. Co-inventor with Dr. S. Shringi, Manipal University Jaipur.

Education

Manipal University Jaipur — B.Tech, Computer Science

Sep 2022 – May 2026

CGPA 8.16/10 | Coursework: Deep Learning, Digital Signal Processing

The Espee Global School, Roorkee — Class XII (PCM) 95%, Class X 98%

Apr 2019 – Mar 2022

Leadership

Convener, Startup Weekend Jaipur (Google for Startups)

Apr 2024 – Nov 2024

Led a 150-member organising team for a 54-hour hackathon with 160+ participants.